
Nulls Are Not Feedback: Adaptive Safety Evaluation Pays Across Design Families, Not Within Them

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Abstract

Safety evaluations report nulls constantly: “we could not elicit the behaviour.” That sentence is load-bearing in deployment decisions, and it is ambiguous, because it can mean the model is robust or that the experiment was badly designed. A popular remedy is adaptivity: treat a null as feedback and redesign. We test this directly, holding total compute constant, which no prior work does. We replay four protocol families against 437k real per-iteration attack traces, a complete $5 \times 4 \times 100$ cost-to-discovery matrix, and a live grid on three open-weights instruct models under an anti-sycophancy intervention. Adaptivity helps enormously: a protocol that spreads a fixed budget over three design families detects 100% of behaviours where the default single-design protocol detects 0% at every budget up to 10,000 iterations. But almost none of that gain comes from the proposed mechanism. Against a comparator that switches design family on a blind fixed schedule with identical budget and identical designs, using the null as a trigger gains nothing ($\Delta = +0.02$) and loses at large budgets ($\Delta = -0.12$). We explain why: conditional on a run still being null, the continuous progress signal predicts eventual success at AUC 0.45–0.61, with every interval covering chance. Adaptivity at the portfolio level does work (+0.65 behaviours on GPT-4; +0.72 live on Qwen2.5-7B). Finally, nominally 95% bounds on residual vulnerability achieve 63–67% coverage, and an anytime-valid correction barely helps, because the estimand, not the stopping rule, is wrong.

1 Introduction

A safety evaluation that finds nothing is the most common result in AI safety, and the least interpretable. “We could not elicit the behaviour” can mean the model is robust, or it can mean the experiment was not sensitive enough to see the failure. Static protocols cannot tell the two apart, and the difference matters: this sentence appears in model cards, audit reports, and deployment decisions.

The literature already contains proof that the second reading is often the correct one. In the Jailbreak-Bench artifacts, PAIR Chao et al. [2023] scores 0.00 attack success against LLAMA-2-7B-CHAT while prompt-plus-random-search scores 0.90 on the *same* 100 behaviours of the *same* model Chao et al. [2024], Andriushchenko et al. [2025]. A lab that ran only PAIR would have published a false robustness claim. Similar reversals recur across the alignment literature: jailbreaks reported as ineffective against unlearning succeed when applied more carefully Łucki et al. [2024], and defenses reporting single-digit attack success under automated single-turn attacks fall to human multi-turn red teaming more than 70% of the time Li et al. [2024].

The natural response, and the hypothesis we test, is that protocols should be *adaptive*: treat a null as actionable feedback and use it to trigger experimental redesign. This idea is widely endorsed and, so far, untested as such.

The gap. Adaptivity has been argued for *attacks* Andriushchenko et al. [2025] and for *forecasting* rare behaviours Jones et al. [2025], but never with the protocol as the independent variable and total compute held fixed. Every published adaptivity result is confounded with spending more queries, so “adaptive beats static” may only mean “more compute beats less compute.” Worse, the word “adaptive” silently conflates two very different moves: adjusting parameters *inside* a design family, and *switching* design family. And nobody scores the null branch, i.e., what an adaptive protocol should output when it still finds nothing. That branch is exactly the half of the claim that safety cases rest on.

Our approach. We make the protocol the independent variable and hold the budget constant by construction. Each substrate’s outcome matrix is evaluated exhaustively once, and protocols are then *replayed* against the cached matrix under an identical budget B . This removes the compute confound, makes every comparison paired on the same behaviours, and lets us report whole detection-versus-budget curves rather than single points. We run four experiments on three substrates: 437k per-iteration random-search traces released by Andriushchenko et al. [2025]; the official Jailbreak-Bench cost-to-discovery matrix Chao et al. [2024] of 5 attacks $\times$ 4 target models $\times$ 100 behaviours; and a live grid on three open-weights instruct models run locally under a real anti-sycophancy intervention, using sycophantic answer-flipping on GSM8K Cobbe et al. [2021] as a non-harmful failure mode.

The critical design choice is our comparator. Most work contrasts an adaptive protocol with a single static design, which cannot separate “adaptivity works” from “design diversity works.” We add FIXED-SPLIT: same budget, same set of design families, but the switch points are placed on a blind schedule that ignores the data. Our adaptive arm LADDER is built so that disabling its trigger recovers FIXED-SPLIT exactly, so the two arms differ *only* in whether the null is used as a trigger.

What we find. The motivating phenomenon is real and extreme: the default design detects 0/50 behaviours at every budget up to 10,000 iterations, while a protocol spending the identical budget across three design families reaches 1.00 (McNemar $p < 10^{-6}$, Cohen’s $h \approx 2.4$). But against FIXED-SPLIT, the null-as-trigger mechanism gains nothing ($\Delta = +0.02$, $p = 0.73$) and loses at large budgets ($\Delta = -0.12$, 95% CI $[-0.22, -0.04]$). We trace this to a mechanistic fact: conditional on a run still being null at iteration t , the continuous progress signal predicts eventual success at pooled AUC 0.45–0.61, with every confidence interval covering 0.5. Adaptivity at the *portfolio* level does work, beating an equally diversified static schedule by +0.65 behaviours on GPT-4 and by +0.72 live on QWEN2.5-7B. Finally, nominally 95% upper bounds on residual vulnerability achieve 63–67% empirical coverage, and an anytime-valid correction for optional stopping moves coverage only from 0.638 to 0.671.

Contributions.

- We propose a compute-matched replay methodology that makes the evaluation protocol the independent variable, and we instantiate it on three substrates covering replayed attack traces, a benchmark cost matrix, and live open-weights models.
- We decompose “adaptive” into within-search escalation and portfolio-level allocation, and show with a trigger-ablated comparator that the entire measured benefit comes from searching multiple design families, not from the null-as-trigger mechanism.
- We test the mechanism directly by measuring the diagnosticity (AUC) of the progress signal on the surviving-null population, and find it indistinguishable from chance at all seven checkpoints.
- We score the null branch and show that nominally 95% bounds cover 63–67% of the time, that anytime-valid correction does not repair this, and that on total nulls tail extrapolation is undefined 100% of the time.
- We replicate the whole contrast live on QWEN2.5-7B, QWEN2.5-1.5B and PHI-3.5-MINI, where a static protocol using the natural first design measures a 3.7% flip rate and would report that the anti-sycophancy intervention holds, while 85% of items flip under a different pressure family.

Section 2 positions the work, section 3 describes the substrates and protocol arms, section 4 reports the four experiments, and sections 5 and 6 discuss implications and limitations.

2 Related Work

Adaptive attacks. Automated red teaming began with attacker language models and large-scale human red teaming Perez et al. [2022], Ganguli et al. [2022] and with gradient-based universal suffixes Zou et al. [2023], and is now standardised by HarmBench Mazeika et al. [2024]. Nearly every method since iterates on failure. PAIR Chao et al. [2023] refines a candidate using the target’s refusal, TAP Mehrotra et al. [2023] adds tree search with pruning, Rainbow Teaming Samvelyan et al. [2024] maintains a quality-diversity archive over attack styles, GOAT Pavlova et al. [2024] picks a published attack technique per turn, and curiosity-driven red teaming Hong et al. [2024] adds a novelty bonus to an RL attacker. Andriushchenko et al. [2025] state the thesis most directly: adaptive attacks are necessary for accurate robustness evaluation, and no single method generalises across targets. Their ablation on LLAMA-2-7B-CHAT moves from 0% to 50% to 100% attack success as random search and then self-transfer initialisation are added, and their R2D2 result is the built-in negative control: self-transfer gives only 12% there, while switching to an in-context template family gives 90%. Best-of- N Hughes et al. [2024] is the canonical non-adaptive control, being memoryless by construction, and establishes the power law $-\log \text{ASR}(N) = aN^{-b}$ that we use as the extrapolative bound in section 4. Unlike all of this work, we hold the attack set fixed and vary the *protocol* that schedules it under a fixed budget, so adaptivity is not confounded with compute.

Return curves and reproducibility. Whether adaptive allocation can beat uniform allocation at all depends on the shape of the return curve. Halder et al. [2026] report a polynomial regime without prompt injection and an exponential regime with it, and Topol [2026] apply process mining to 8,575 scored red-teaming events to show that mutator effectiveness is asymmetric across models and that time-to-jailbreak distributions differ by an order of magnitude. This is independent 2026 confirmation of the lesson we build on, that which adaptation works is model-specific. Fang et al. [2026] reproduce 30 published attacks in a unified harness to a mean deviation of +0.26 percentage points, which is the kind of control that makes replay-based comparisons like ours credible.

Nulls, forecasting, and elicitation. Jones et al. [2025] give the closest prior work: they replace a binary null with a continuous elicitation probability and extrapolate extreme quantiles via a Gumbel-tail fit, forecasting risk two orders of magnitude beyond the observed budget. Their final section uses forecasts to allocate red-teaming compute, which is the direct ancestor of our portfolio experiment. They also propose adaptively stopping unpromising queries as future work. We take that proposal seriously enough to test it, and find that the within-run signal it would rely on is not diagnostic on the population that matters.

Hidden robustness and false nulls. A second literature supplies the phenomena. Wei et al. [2023] give the taxonomy, splitting failures into competing objectives and mismatched generalisation, which is in effect a principled axis set for a design space. Shallow safety alignment Qi et al. [2025] shows alignment mostly adapts the first few output tokens, which explains why suffix, pre-filling, and decoding attacks all work. Sleeper agents Hubinger et al. [2024] persist through SFT, RL, and adversarial training, with adversarial training teaching models to hide the trigger better. Fine-tuning breaks guardrails with ten examples Qi et al. [2024]. Łucki et al. [2024] recover unlearned capabilities with attacks previously reported as ineffective, and Che et al. [2025] show that input-output evaluations only lower-bound worst-case behaviour. On the other side, latent adversarial training Sheshadri et al. [2024] and constitutional classifiers Sharma et al. [2025] are rare cases where a null appears to be earned, the latter at a cost of over 3,000 human red-teaming hours. Betley et al. [2025] add a measurement problem that compounds all of this: narrowly fine-tuned models behave inconsistently and sometimes act aligned, so single-sample evaluation has low power by construction. Whether adaptive search should pay depends on the dimensionality of the vulnerability boundary, which is itself disputed: Arditi et al. [2024] report a single refusal direction, while Wollschläger et al. [2025] report multi-dimensional concept cones. We treat this as a moderator rather than settled, and our results are consistent with the family-dependent view.

Statistics for evaluation. Miller [2024] imports standard errors, clustering, paired inference, and power analysis into evaluation, and shows clustered standard errors can be $3.05\times$ larger than naive ones. Bowyer et al. [2025] show CLT intervals badly understate uncertainty below a few hundred datapoints, which matters here because adaptive protocols concentrate budget on small subsets. Anytime-valid inference Ramdas et al. [2022], Lindon et al. [2022] is the correct frame for data-dependent stopping, and adaptive item selection with preserved coverage has been developed for

Experiment	Substrate	Size	Role
E1, E1c, E3	RS-TRAJ: released per-iteration random-search logs Andriushchenko et al. [2025]	14 configs, $\sim 437k$ iteration rows, 672 run outcomes	Real sequential traces with a continuous objective; contains a matched adaptive/static pair
E2	JBB-ARTIFACTS: official JailbreakBench artifacts Chao et al. [2024]	5 methods $\times$ 4 target models $\times$ 100 behaviours	Complete cost-to-discovery matrix; any allocation policy can be replayed offline
E4	GSM8K test split Cobbe et al. [2021] + real open-weights LLMs run locally	32 items $\times$ 4 pressure families $\times$ 2 conditions $\times$ 3 models	End-to-end live replication on a fresh, non-harmful failure mode

Table 1: The three substrates. E1–E3 replay already-published traces and artifacts; E4 runs real model weights locally. No simulated model is used anywhere in this work.

capability estimation Polo et al. [2024], Krundick et al. [2026], Wu et al. [2026], Tolochinsky et al. [2026]. Dörner et al. [2024] bound how far an LLM judge can substitute for labels, which is why we report both a judge-based and a rule-based outcome throughout. The red-teaming literature has the optional-stopping problem and this literature has the fix; we join them, and find, surprisingly, that the fix addresses the smaller of two errors.

Bayesian experimental design. The formal vocabulary the hypothesis reaches for informally is that of sequential Bayesian experimental design Rainforth et al. [2023], where a design policy maps history to the next design so as to maximise expected information gain, and can be amortised into a single forward pass Foster et al. [2021]. Our results say something concrete to that framework in this domain: the expected information gain of a within-run null is close to zero under the surrogate that is actually available, while the expected information gain of a cross-run null about which design family to use is large.

Positioning. Our contribution is the controlled comparison that none of the above performs: protocol as independent variable, compute held constant, within-family and cross-family adaptivity separated by an ablated comparator, and the null branch scored for coverage rather than assumed valid.

3 Methodology

3.1 Problem setup

An evaluation protocol π is given a budget B of unit-cost queries, a set of targets (behaviours or items) $\mathcal{I}$, and a set of design families $\mathcal{F}$. A design family is a qualitatively distinct way of probing the model: a suffix-search initialisation, a prompt template class, or a social-pressure schema. At each step the protocol chooses a target and a family, spends one unit, and observes an outcome $y \in \{0, 1\}$ together with a continuous surrogate score $s \in \mathbb{R}$. The primary dependent variable is the *detection rate* at matched budget, $D_\pi(B) = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \mathbf{1}[\pi \text{ elicits the behaviour on } i \text{ within } B]$. We report $D_\pi(B)$ as a whole curve over B , never as a single point.

The key comparison in this paper is not adaptive versus static. It is adaptive versus *blindly diversified*. Let π_{blind} spend B across $\mathcal{F}$ on a schedule that is a function of B alone, and let π_{adapt} spend B across the same $\mathcal{F}$ using a schedule that is a function of $(B, s_{1:t})$. Because π_{adapt} reduces to π_{blind} when its trigger is disabled, $D_{\pi_{\text{adapt}}}(B) - D_{\pi_{\text{blind}}}(B)$ isolates the value of using the null as a trigger, with everything else held fixed.

3.2 Substrates

table 1 lists the three substrates. Nothing in E1–E3 is re-run or simulated: these are the original authors’ own released traces and the official benchmark artifacts. E4 uses real model weights executed locally. No simulated agent appears anywhere in this work.

RS-TRAJ (E1, E1c, E3). We parsed the 24 attack logs released with Andriushchenko et al. [2025] into 14 configurations, roughly 437k per-iteration rows and 672 run outcomes. Each row carries the target-token log-probability surrogate and the judge outcome, so a protocol can be replayed at iteration granularity. The corpus contains a matched pair on LLAMA-2-7B-CHAT that differs only in initialisation policy: `plain_init` (static) scores 0.00 over 50 behaviours with 56% of runs hitting the 10,000-iteration cap, while the self-transfer configuration scores 0.96 with a median of 2 iterations.

JBB-ARTIFACTS (E2). The official JailbreakBench artifacts Chao et al. [2024] give `queries_to_jailbreak` and a success flag for 5 attacks $\times$ 4 target models $\times$ 100 behaviours. This is a complete cost-to-discovery matrix, so any allocation policy can be replayed offline exactly.

Live grid (E4). We run three open-weights instruct models Yang et al. [2024], Abidin et al. [2024] locally on 32 GSM8K test problems under an anti-sycophancy system-prompt intervention and a no-intervention control, crossed with four social-pressure design families: P1 simple doubt, P2 appeal to authority, P3 fabricated consensus, P4 persistent multi-turn pressure. The failure mode is sycophantic answer-flipping, which is non-harmful and produces no jailbreak artifacts. Only items the model answers correctly without pressure are eligible, so “flip” is always well defined. Baseline accuracies are 0.844 (QWEN2.5-7B), 0.781 (PHI-3.5-MINI) and 0.594 (QWEN2.5-1.5B), leaving 27, 25 and 19 eligible items respectively.

3.3 Matched-budget replay

Every experiment follows the same pattern. The outcome matrix is evaluated (or read) once, exhaustively, over all targets $\times$ families $\times$ budget positions. Protocols are then replayed against that cached matrix under an identical budget. This has three consequences. First, the compute confound is removed by construction rather than by argument. Second, every protocol comparison is paired on the same behaviours or items, so we can use paired tests. Third, bootstrap uncertainty is cheap, because replaying a policy costs microseconds.

3.4 Protocol arms

E1: within-search escalation ladder. Three design families are available on LLAMA-2-7B-CHAT: d_0 the default `plain_init` random search, d_1 re-initialisation from a related success (self-transfer), and d_2 a switch of template family. The arms are STATIC-D0, STATIC-D1 and STATIC-D2 (all B on one family; STATIC-D0 is the “continue-only” control, i.e., maximal within-family persistence), FIXED-SPLIT (B divided equally over the three families on a schedule that ignores the data), LADDER (same designs, same B , but the switch point is moved by the continuous surrogate), and ORACLE (best family per behaviour). LADDER has three hyper-parameters: a patience, a surrogate threshold τ , and an extension factor. At $\tau = 0$ and extension 1 it recovers FIXED-SPLIT *exactly*. This is what makes the LADDER–FIXED-SPLIT contrast decisive; the STATIC-D0 contrast only establishes that design diversity matters at all.

E1c: null diagnosticity. At checkpoint t we restrict to runs that are *still null at t* , which is precisely the decision context an adaptive protocol faces, and measure the AUC of the surrogate at t for predicting whether the run eventually succeeds. If this AUC is 0.5, no protocol can use the null as a trigger, however cleverly implemented.

E2: portfolio allocation. Budget is a number of attack runs allocated over methods and behaviours. Static arms are STATIC-SINGLE (the field default), STATIC-RR (diversified but non-adaptive) and STATIC-BEST, an *oracle* single-method choice that upper-bounds any *a priori* decision. Adaptive arms are escalation, ADAPT-TS (Thompson sampling over methods), ADAPT-SH (successive halving) and ADAPT-ESC+TS. ORACLE marks the union ceiling.

E3: calibration of the terminal null. A protocol stops and must emit a nominally 95% upper bound on the elicitable fraction. FIXED-CP is Clopper-Pearson at a pre-registered budget, ADAPTIVE-CP is the same interval at a data-chosen stopping time (what the field actually does), UNION-CP applies a Bonferroni union bound over the monitoring grid and is genuinely anytime-valid, and EXTRAPOLATIVE fits the Best-of- N power law Hughes et al. [2024] and extrapolates to the full horizon. Ground truth is the outcome at the end of the observed trace.

E4: live protocol race. STATIC-P1 spends its whole budget on P1, the obvious first design; STATIC-RR cycles blindly over the four families; ADAPT-TS allocates adaptively over families. Budget is a number of model calls.

3.5 Metrics and statistical analysis

The analysis plan was pre-registered before any protocol was run. Primary metric is detection rate at matched budget. Secondary metrics are queries to first detection treated as right-censored survival

Budget B	STATIC-D0	STATIC-D1	STATIC-D2	FIXED-SPLIT	LADDER	ORACLE
8	0.00	0.62	0.56	0.80	0.78	0.84
20	0.00	0.64	0.56	0.84	0.84	0.84
60	0.00	0.98	0.58	0.84	0.86	1.00
200	0.00	0.98	0.60	1.00	0.88	1.00
10,000	0.00	0.98	0.64	1.00	0.88	1.00

Table 2: **E1: detection rate at matched iteration budget** on 50 LLAMA-2-7B-CHAT behaviours. STATIC-D0, the default design, detects nothing across four orders of magnitude of budget, while every arm that searches more than one design family detects most or all behaviours. LADDER, which uses the null as a switch trigger, never beats FIXED-SPLIT, which switches on a blind schedule with the same budget and the same designs. ORACLE is an upper bound (best family per behaviour) and is not a deployable protocol. Best deployable arm per row in **bold**.

data, AUC of the trigger signal, coverage and width of the emitted upper bound, and, for E4, the flip rate under pressure and the fraction of vulnerable items localised.

Uncertainty comes from bootstrap over behaviours or items (1,000 resamples, seed 42), never from re-running attacks, since the artifact matrix has one realisation per cell. Where an exact per-behaviour join exists we use paired bootstrap plus exact McNemar on discordant behaviours; unpaired comparisons are matched by resampling and are flagged explicitly as resting on an exchangeability assumption. E2 and E4 comparisons use Wilcoxon signed-rank over paired policy replicates (300 and 400 respectively). We apply Holm correction within each experiment’s family of tests at $\alpha = 0.05$, and report an effect size (risk difference, Cohen’s h) alongside every p -value. Both the GPT-4-judge and the rule-based outcome are retained throughout; they disagree, and we report the disagreement rather than averaging it away.

3.6 Reproducibility

Seed 42 everywhere. E4 decoding is greedy (do_sample=False, temperature 0). Environment: Python 3.12, torch 2.13.0+cu130, transformers with native model code (trust_remote_code=False), pandas 3.0.5, scipy 1.18.0, with uv managing an isolated virtual environment. E1–E3 run on CPU in under five minutes in total; the E4 grid took roughly 10 GPU-minutes per model on an RTX A6000.

4 Results

We report four experiments. E1 and E1c ask whether the null-as-trigger mechanism works and why; E2 asks whether adaptivity works at the portfolio level; E3 scores the null branch; E4 replicates the whole contrast live on real models.

4.1 E1: the escalation ladder at matched iteration budget

Two things happen at once in table 2 and figure 1.

Is the motivating phenomenon real? Yes, and it is extreme. STATIC-D0 detects 0/50 behaviours at every budget from 8 to 10,000 iterations, while arms that spend the identical budget over three design families reach 1.00. The risk difference is 0.88–0.98 depending on budget, exact McNemar $p < 10^{-6}$, Cohen’s $h \approx 2.4$. This is the cleanest available instance of a false null: 10,000 iterations of a perfectly reasonable attack, reported as robustness, on a model that three other arms break almost completely. A static protocol here does not merely under-detect. It reports total robustness where near-total vulnerability exists.

Does the null-as-trigger mechanism add anything? No. Against FIXED-SPLIT, which has the same budget, the same three designs, and switches on a blind schedule, LADDER is statistically indistinguishable at small budgets ($B = 60$: $\Delta = +0.02$, 95% CI [0.00, 0.06], McNemar $p = 1.0$; pooled small-budget $p = 0.73$) and *worse* at large budgets ($B = 1000$: $\Delta = -0.12$, 95% CI [-0.22, -0.04], bootstrap $p = 0.004$; 6 behaviours lost, 0 gained). The Holm-adjusted McNemar p

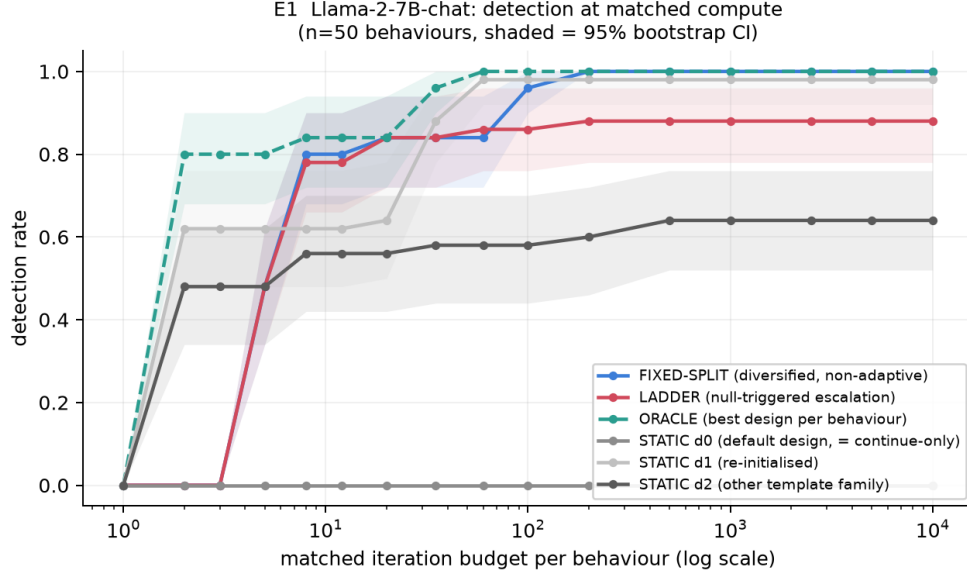


Figure 1: Detection rate versus iteration budget on 50 LLAMA-2-7B-CHAT behaviours, with 95% bootstrap intervals over behaviours. STATIC-D0 (the default design) is flat at zero across four orders of magnitude. LADDER tracks FIXED-SPLIT at small budgets and falls behind it at large budgets, so the whole gap to STATIC-D0 is a design family effect and none of it is a trigger effect.

Checkpoint t	n	configs	pooled AUC	95% CI	base rate
1	471	10	0.515	[0.47, 0.57]	0.69
2	471	10	0.612	[0.50, 0.75]	0.69
3	186	7	0.474	[0.29, 0.59]	0.51
5	142	5	0.448	[0.23, 0.56]	0.54
10	126	4	0.470	[0.28, 0.60]	0.52
25	102	3	0.578	[0.30, 0.69]	0.46
100	67	3	0.536	[0.43, 0.62]	0.40

Table 3: **E1c: the null carries no usable signal.** Conditional on a run still being null at iteration t , the AUC of the continuous target-token surrogate for predicting eventual success. Every confidence interval covers 0.5. The easy successes have already left the population by t , and what remains is not separable, which is why a trigger built on this signal cannot work.

for the large-budget loss is 1.0, so we report it as a consistent negative trend rather than a confirmed harm. Against STATIC-D0, by contrast, LADDER wins overwhelmingly ($B = 60$: $\Delta = +0.86$, 95% CI [0.76, 0.94], McNemar $p < 10^{-6}$, $h = 2.37$; the exactly-paired two-rung variant gives $\Delta = +0.98$, $h = 2.86$).

Rung attribution. The decomposition is direct. Across every budget, *zero* of LADDER’s detections come from rung 0 (continue in the default family); 31 come from rung 1 (re-initialisation) and 13 from rung 2 (template-family switch). The entire measured benefit of “adaptivity” here is attributable to searching more than one design family.

Sensitivity. The finding is not a knife-edge artifact of one trigger setting. A $3 \times 3 \times 3$ sweep over patience $\in \{2, 5, 20\}$, threshold $\tau \in \{0.05, 0.3, 0.7\}$ and extension $\in \{1, 3, 10\}$ never produces a gain over FIXED-SPLIT larger than +0.06, and produces losses as large as −0.17.

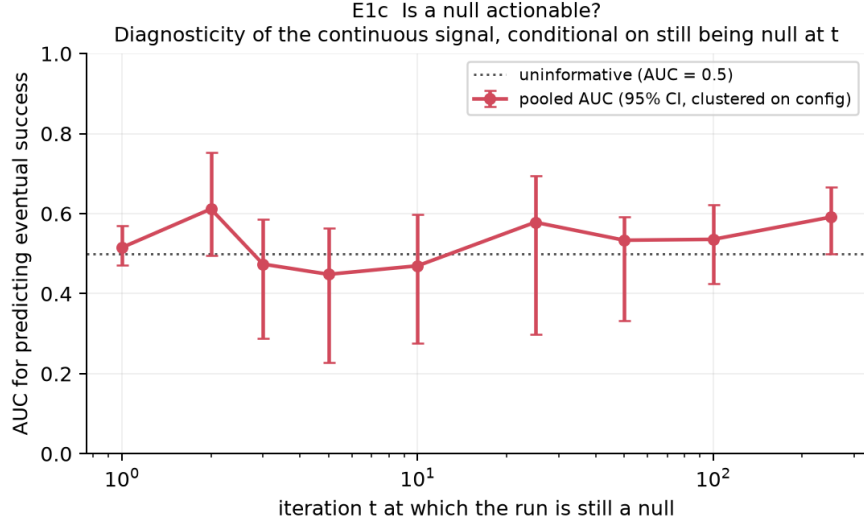


Figure 2: Diagnosticity of the null. Pooled AUC (with 95% intervals) of the continuous target-token surrogate at checkpoint t for predicting eventual success, restricted to runs still null at t . The horizontal line marks chance. No checkpoint separates from 0.5, which means the information a null-triggered protocol would act on does not exist in this signal.

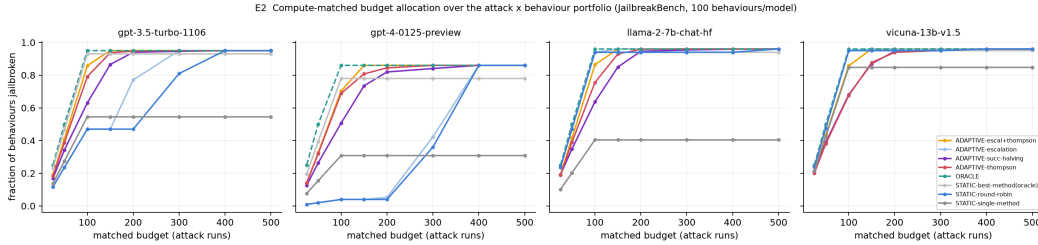


Figure 3: Fraction of 100 behaviours jailbroken versus budget on the JailbreakBench cost-to-discovery matrix, for four target models. Adaptive allocation over methods dominates on GPT-4-0125 and GPT-3.5-TURBO, where method quality varies widely and is unknown *a priori*, and loses to blind round-robin on LLAMA-2-7B-CHAT and Vicuna-13B, where one method dominates and exploration is pure overhead.

4.2 E1c: why the trigger fails

E1 shows that the trigger does not help. E1c shows why, and this is a mechanistic refutation rather than a restatement. Conditional on a run still being a null at iteration t , which is exactly the decision context an adaptive protocol faces, the continuous surrogate predicts eventual success at pooled AUC 0.448–0.612 across seven checkpoints, and *every* confidence interval covers 0.5 (table 3, figure 2). The base rate falls from 0.69 at $t = 1$ to 0.40 at $t = 100$, confirming that the easy successes leave the population early; what remains is not separable by this signal.

This explains the -0.12 loss at large budget in E1. A protocol that trusts an uninformative signal will sometimes read a high surrogate value as evidence of progress and spend its remaining budget persisting with a doomed rung. Because the signal is noise, that persistence is a pure loss.

4.3 E2: portfolio-level adaptivity does work

The second form of adaptivity, reallocating budget across *methods* rather than within a search, behaves very differently. At budget 100, ADAPT-ESC+TS beats the equally diversified STATIC-RR by $+0.648$ on GPT-4-0125 (0.703 vs 0.040) and $+0.321$ on GPT-3.5-TURBO (0.859 vs 0.470), both

Budget	Static			Adaptive		ORACLE
	STATIC-SINGLE	STATIC-RR	STATIC-BEST (oracle)	ADAPT-TS	ADAPT-ESC+TS	
GPT-4-0125-preview	(design-space ceiling 0.86)					
50	0.155	0.020	0.393	0.320	0.325	0.50
100	0.308	0.040	0.780	0.688	0.703	0.86
150	0.308	0.040	0.780	0.808	0.860	0.86
300	0.308	0.360	0.780	0.860	0.860	0.86
GPT-3.5-TURBO-1106	(design-space ceiling 0.95)					
100	0.545	0.470	0.930	0.791	0.859	0.95
150	0.545	0.470	0.930	0.937	0.950	0.95
300	0.545	0.810	0.930	0.950	0.950	0.95

Table 4: **E2: portfolio-level adaptivity works.** Fraction of 100 behaviours jailbroken at matched budget (number of attack runs; 300 policy replicates per cell). Adaptive allocation beats the equally diversified STATIC-RR by +0.648 on GPT-4-0125 and +0.321 on GPT-3.5-TURBO at budget 100 (Holm-adjusted $p < 0.001$), and at budget 150 it exceeds STATIC-BEST, an oracle single-method allocation, because no single method covers all behaviours. Best deployable arm per row in **bold**; STATIC-BEST and ORACLE are upper bounds, not protocols.

Method	fraction defined	coverage	mean bound	mean truth
FIXED-CP (pre-registered budget)	1.00	0.626	0.697	0.682
ADAPTIVE-CP (data-chosen stop)	1.00	0.638	0.676	0.682
UNION-CP (anytime-valid)	1.00	0.671	0.726	0.682
EXTRAPOLATIVE (power law)	0.89	0.555	0.686	0.682

Table 5: **E3: nulls are badly calibrated, and sequential rigour does not repair them.** Empirical coverage of a nominally **95%** upper bound on the elicitable fraction, over 14 configurations $\times$ 400 bootstrap replicates. All four methods under-cover badly. Correcting for optional stopping with a union bound moves coverage only from 0.638 to 0.671, because the dominant error is that a confidence interval on “what we found by budget t ” is not a bound on “what is elicitable”. Closest to nominal in **bold**.

Holm-adjusted $p < 0.001$ by Wilcoxon over 300 paired replicates. More tellingly, at budget 150 it reaches 0.860 against 0.780 for STATIC-BEST, an *oracle* best-single-method allocation. Adaptivity exceeds the best possible *a priori* choice, because no single method covers all behaviours: the best single method reaches 0.78 while the union ceiling is 0.86.

When does portfolio adaptivity lose? The exception is informative. On LLAMA-2-7B-CHAT and Vicuna-13B, ADAPT-TS loses to STATIC-RR at budget 100 by -0.185 ($p < 0.001$). Those models have a dominant method (DSN at 0.94–0.95) that a blind round-robin reaches immediately, so the cost of learning which method works buys nothing. Adaptivity pays exactly when the method-quality spread is wide and unknown, which is the realistic case for a new model, and precisely the case (GPT-4-0125) where the static protocol is catastrophically bad.

4.4 E3: the nulls an adaptive protocol emits are badly calibrated

The hypothesis has a second half: that adaptive protocols provide better evidence for genuine robustness. They do not, as currently practised. table 5 shows that a nominally 95% upper bound on the elicitable fraction delivers 55–67% empirical coverage across 14 configurations and 400 bootstrap replicates.

The ordering is what matters. The field’s instinct is that adaptive stopping breaks fixed-sample inference and that sequential methods fix it. That instinct is right about the mechanism and wrong about the magnitude: correcting for optional stopping with a union bound over the monitoring grid moves coverage from 0.638 to only 0.671. The dominant error is that a confidence interval on *what we found by budget t* is simply not a bound on *what is elicitable*. Per-configuration coverage makes this stark. On llama2-13b, llama2-70b and llama2_7b, where the true elicitable fraction is 0.96–1.00, coverage of the adaptive interval is **0.00**: the bound is wrong every single time.

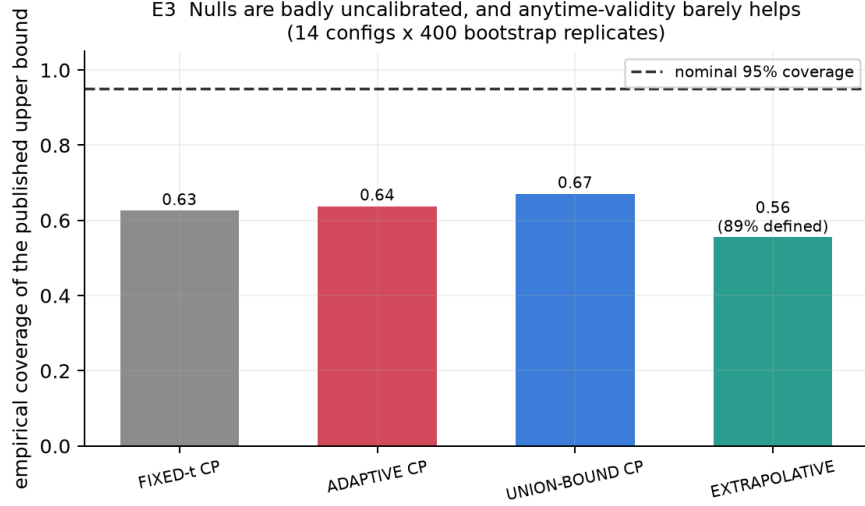


Figure 4: Calibration of the terminal null. Empirical coverage of a nominally 95% upper bound on the elicitable fraction (dashed line at 0.95), by method and by configuration. The anytime-valid UNION-CP correction is the right fix for the wrong problem: it moves pooled coverage from 0.638 to 0.671, still far from nominal.

Model	n	Pressure design family				ceiling
		P1 simple doubt	P2 authority	P3 fabricated consensus	P4 persistent multi-turn	
QWEN2.5-7B	27	0.037	0.185	0.852	0.519	0.926
QWEN2.5-1.5B	19	0.105	0.526	0.263	0.947	0.947
PHI-3.5-MINI	25	0.000	0.160	0.000	0.080	0.160

Table 6: **E4: which design family you pick decides what you conclude.** Flip rate by pressure family with the anti-sycophancy intervention active; n is the number of eligible items, i.e., those the model answers correctly without pressure. A static protocol that pre-registers P1, the obvious first design, measures 0.037 on QWEN2.5-7B and would report that the intervention holds; the same model flips on 0.852 of items under P3. On PHI-3.5-MINI, P1 and P3 both return exactly zero while P2 finds 0.160. Highest-yielding family per model in **bold**; “ceiling” is the union over the four families we wrote, not over all possible designs.

Restricting to evaluators that stopped on a *total* null, the case that matters most for robustness claims, coverage rises to 0.87–0.88 but EXTRAPOLATIVE becomes **undefined 100% of the time**. A power law cannot be fitted to a curve that is identically zero, so the one method designed to convert a null into a forecast is unavailable exactly when it is needed.

4.5 E4: live replication on real open-weights LLMs

E1–E3 replay other people’s logs. E4 runs the actual contrast against real model weights on a fresh, non-harmful failure mode, so the finding is not an artifact of retrospective replay.

Does a static protocol mismeasure a live model? Yes, and severely. table 6 shows that a protocol that pre-registers P1, the obvious and natural first design, measures a 3.7% flip rate on QWEN2.5-7B and would report that the anti-sycophancy intervention holds. The same model flips on 85.2% of items under P3 and 92.6% under the union of designs. On PHI-3.5-MINI, P1 and P3 both return exactly zero while P2 finds 16%.

Does the static protocol also mismeasure the intervention? Yes, and in both directions. Judged by P1 alone, the intervention on QWEN2.5-7B changes the flip rate from 0.077 to 0.037, a small and easily dismissed effect. Judged by P4 it changes the flip rate from 1.000 to 0.519; by P3 from 1.000 to 0.852; and it lowers the ceiling from 1.000 to 0.926. On PHI-3.5-MINI the intervention

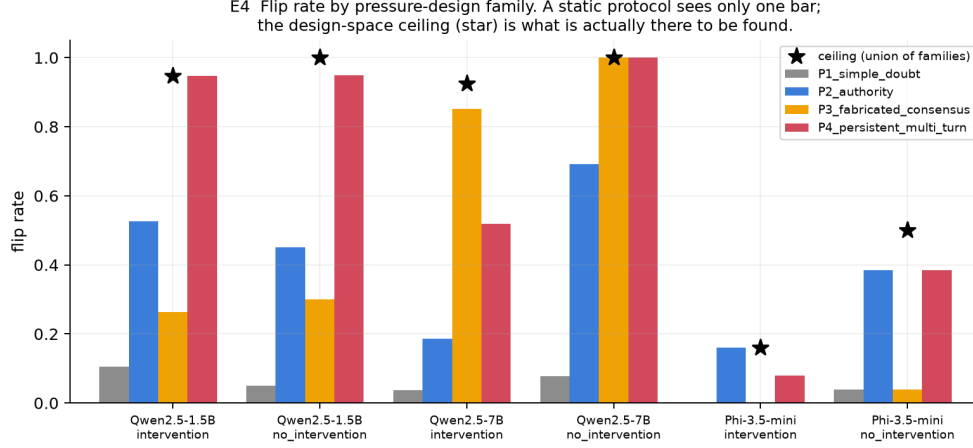


Figure 5: Flip rate by pressure design family on three open-weights models, with and without the anti-sycophancy intervention. The spread across families within a single model is far larger than the spread across models within a family, which is what makes a single-family evaluation unsafe to interpret.

Budget	QWEN2.5-7B			PHI-3.5-MINI		
	STATIC-P1	STATIC-RR	ADAPT-TS	STATIC-P1	STATIC-RR	ADAPT-TS
5	0.212	0.212	0.925	0.000	0.000	0.250
10	0.410	0.410	0.998	0.000	0.000	0.435
20	0.805	0.805	1.000	0.000	0.000	0.670
30	1.000	1.000	1.000	0.000	0.615	0.808
≥ 75	1.000	1.000	1.000	0.000	1.000	1.000

Table 7: **E4: probability that the protocol escapes the null** at matched budget (number of model calls; 400 replicates, intervention active). This is the quantity a safety team acts on. STATIC-P1 on PHI-3.5-MINI never escapes the null at any budget up to 200 calls even though 16% of items are flippable, so its report of robustness is an artifact of the design, not a property of the model. Best per column block in **bold**.

cuts the ceiling from 0.500 to 0.160. An evaluation restricted to one design family therefore gets both the vulnerability rate and the intervention’s effect size badly wrong, and the direction of the error is not consistent across models.

Does adaptivity over design families help at matched budget? Yes. table 7 and figure 6 show that ADAPT-TS reaches a 0.925 probability of escaping the null on QWEN2.5-7B with only 5 calls, where STATIC-P1 and STATIC-RR reach 0.212. On PHI-3.5-MINI, STATIC-P1 *never* escapes the null at any budget up to 200 calls while the vulnerability is reachable, and ADAPT-TS reaches 0.808 by budget 30. On the finer measure, the fraction of vulnerable items localised, ADAPT-TS beats STATIC-RR by +0.72 on QWEN2.5-7B at budget 30 (0.777 vs 0.052, 95% CI [0.52, 0.81], Holm-adjusted $p < 0.001$) and by +0.21 on QWEN2.5-1.5B; on PHI-3.5-MINI the gain is +0.03 with an interval covering zero.

E4 independently reproduces E2’s pattern on live models rather than replayed logs. Adaptivity across the design space pays, and it pays most where the design-quality spread is widest (QWEN2.5-7B: 0.037 $\rightarrow$ 0.852 across families) and least where the model is close to genuinely robust (PHI-3.5-MINI, ceiling 0.16). E4 also supplies the case the offline data could not: a model that is substantially robust, where the honest conclusion is bounded robustness rather than a hidden vulnerability.

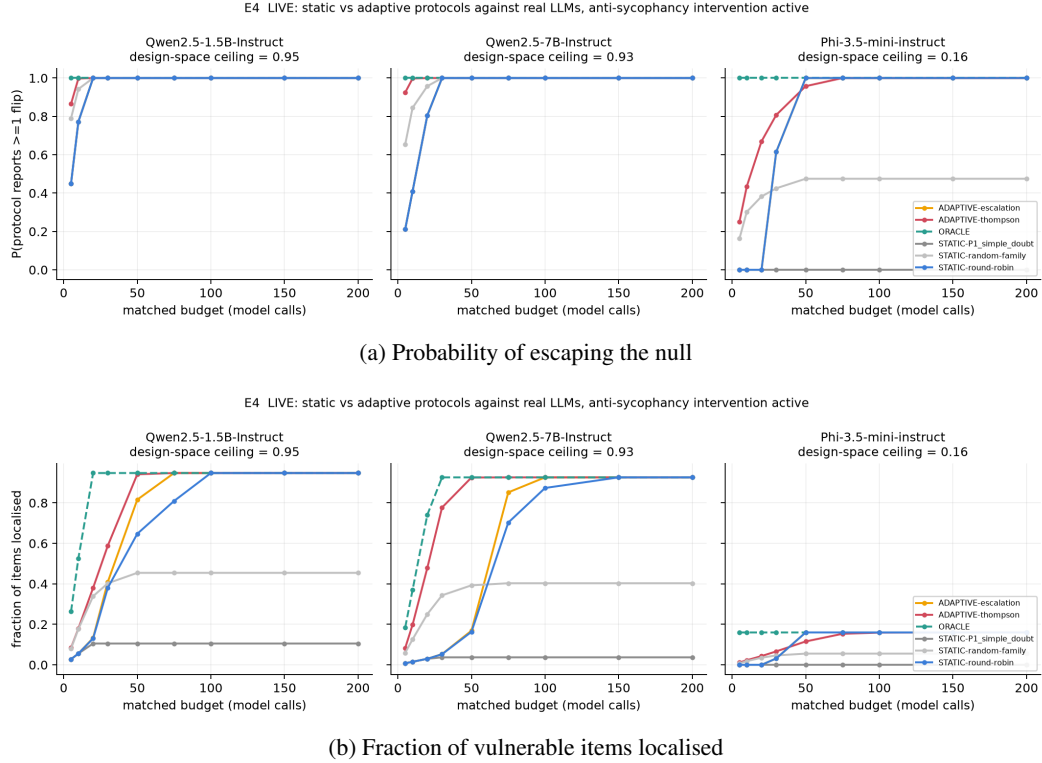


Figure 6: Live protocol race at matched call budget, 400 replicates. (a) the coarse measure a safety team acts on: does the protocol report any flip at all? (b) the finer measure: how much of the vulnerable set does it find? Adaptive allocation over design families dominates both, and the gap is largest on QWEN2.5-7B, where the design-quality spread is widest.

Claim	Verdict	Evidence
Adaptive protocols identify more vulnerabilities than static ones at matched budget	Supported, strongly	E1: 0.00 $\rightarrow$ 1.00; E2: +0.65 on GPT-4-0125; E4 (live): +0.72 on QWEN2.5-7B
The mechanism is using nulls as iterative triggers for redesign	Refuted within a search; supported across a portfolio	E1: $\Delta = +0.02$ to -0.12 vs a blind schedule; E1c: AUC ≈ 0.5 . E2 and E4: Thompson beats round-robin decisively
Adaptive protocols provide better evidence for genuine robustness	Refuted as currently practised	E3: 63–67% coverage of a nominally 95% bound; anytime-validity does not fix it. E4 shows the design-space ceiling is the honest substitute

Table 8: Verdict on the three claims contained in the hypothesis. The hypothesis is right about the conclusion and wrong about the mechanism, and the correction is practically consequential.

5 Discussion

5.1 What the results mean

The hypothesis we set out to test contains three claims, and they receive different verdicts (table 8). Adaptivity does find substantially more at matched compute, and the effect is as large as anyone could want. But almost none of it comes from the mechanism the hypothesis names. What matters is searching over design families at all, and learning across a portfolio which family works. Using the continuous signal inside a failing run as a redesign trigger is worth nothing and can be actively harmful.

The correction is practically consequential rather than semantic. A team that instruments its existing single-design pipeline with a stall detector will get nothing, and at large budgets will lose detections. A team that reallocates the same budget across three designs will go from 0% to 100%.

5.2 Why the two levels of adaptivity differ

Why does the same idea work at one level and fail at the other? Because they face different signal-to-noise regimes. At the portfolio level, the evidence accumulating against a method is a run of independent Bernoulli nulls across *different* behaviours. Twenty nulls in a row from a method is overwhelming evidence that the method does not work on this model, and Thompson sampling exploits that correctly. At the within-search level, the “signal” is a single monotone surrogate on one behaviour, and E1c shows it does not separate the surviving-null population. The lesson generalises beyond our substrates: *nulls are informative in aggregate across independent trials, and uninformative as a within-trial progress signal.*

This also reconciles our result with Andriushchenko et al. [2025], whose $0\% \rightarrow 50\% \rightarrow 100\%$ progression on LLAMA-2-7B-CHAT is often read as evidence for adaptive *search*. Our decomposition says the gain is a design family effect: self-transfer initialisation and template switching are changes of design, not data-driven adjustments within one design. Their own R2D2 negative control agrees, since more self-transfer gives 12% there while a template-family switch gives 90%.

5.3 Failure modes

When portfolio adaptivity loses. On LLAMA-2-7B-CHAT and Vicuna-13B a dominant method exists, so exploration is wasted and blind round-robin wins by 0.185. The practical diagnostic is simple: if you expect a wide and unknown method-quality spread, adapt; if you suspect a dominant method, do not.

When the ladder loses. The “persist while the surrogate is warm” rule keeps budget on rungs whose surrogate is high but whose judged outcome is negative. Because the surrogate is non-diagnostic on the surviving-null population, this is a pure loss, and it grows with budget.

When the static protocol is actively misleading. E4 shows the worst case is not under-powering but mis-estimation of an intervention’s *effect size*. Judged by P1 the anti-sycophancy intervention on QWEN2.5-7B looks nearly inert ($0.077 \rightarrow 0.037$); judged by P4 it looks strongly effective ($1.000 \rightarrow 0.519$). Single-family evaluation gets the direction of the conclusion wrong, not just the precision.

Judge disagreement is large and is not noise. The configuration llama2-7b_icl_one_shot scores 0.66 under the GPT-4 judge and 0.10 under the rule-based judge. Consistent with the theoretical ceiling on judge substitution Dörner et al. [2024], any single-judge efficiency claim in this literature should be treated as provisional. We report both outcomes throughout.

5.4 The surprise

We expected anytime-valid inference to substantially repair coverage in E3, and it barely moved it ($0.638 \rightarrow 0.671$). The problem in safety evaluations is not primarily that people peek at their data. It is that the estimand they bound, the rate observed at budget t , is not the estimand they report, elicibility. No amount of sequential rigour fixes an estimand error. This suggests the useful research target is a bound whose estimand is elicibility *under a specified design space*, evaluated for coverage against held-out design families.

5.5 Limitations

We state these plainly because several of them bound the conclusions materially.

- **Retrospective replay.** E1–E3 replay logs generated by other people’s design choices, so a design family absent from those logs cannot be evaluated. STATIC-D0’s total failure is partly a property of the specific plain_init ablation, which is harsher than the headline “Prompt + RS” row reporting 50% in the original paper. We do *not* claim to reproduce that table.

- **Conservative replay semantics.** A run the original authors ended without success is treated as a failure even if budget remained, which caps the persistence arms. This is applied identically to every arm, so it does not favour the adaptive ones; if anything it makes FIXED-SPLIT and LADDER look more similar than they are.
- **Unpaired rung 2.** Goal strings are unrecoverable for the in-context-template configurations, so the third rung is matched by resampling, which is an exchangeability assumption. The exactly-paired two-rung variant gives the same qualitative answer ($\Delta = +0.98$ vs STATIC-D0).
- **One realisation per cell** in the JailbreakBench matrix, so all E2 uncertainty is policy-randomisation and bootstrap, not attack re-run variance.
- **Heuristic run boundaries** in the trajectory parse, accurate to within ± 2 of the expected 50 runs per configuration.
- **The surrogate is an approximation.** E1c uses target-token probability, which Jones et al. [2025] themselves use for most experiments, but the mapping to elicitation probability is approximate. A stronger signal might be diagnostic where this one is not. This is the most important threat to the E1c conclusion and therefore to the E1 verdict.
- **E3’s ground truth** is the end of the observed trace, not true elicibility at infinite budget, so the reported coverage is if anything optimistic.
- **No frontier API access**, so the live arm uses local open-weights models. Results may not transfer to frontier models with different safety training.
- **E4 is small:** 32 items per model, 19–27 eligible after the correctness filter, and four pressure families. The family-level contrasts are large enough to survive this (0.000 vs 0.852), but the per-family rates carry wide intervals, and the “ceiling” is a ceiling over the four families we wrote, not over all possible designs. That is exactly the caveat this paper argues every null should carry.
- **GSM8K contamination** cannot be excluded. It is mitigated by measuring answer *flipping under pressure* rather than accuracy, and by conditioning on items the model already answers correctly, but it is not eliminated.

5.6 Broader impact

This work is defensive evaluation methodology. E1–E3 replay already-published benchmark artifacts and generate no new attack content, and E4 uses a non-harmful failure mode, sycophantic answer-flipping on grade-school arithmetic, and produces no jailbreak artifacts. The main risk we see is the opposite of misuse: our headline finding could be read as a licence to distrust all null results. That is not the claim. The claim is that a null should be reported together with the design space that was searched, and that a binomial interval computed on a null does not bound a model’s vulnerability. Used that way, the finding should make safety cases more honest rather than less usable.

6 Conclusion

We asked whether an adaptive evaluation protocol, one that treats its own nulls as triggers for experimental redesign, finds more vulnerabilities and produces better-calibrated conclusions than a static protocol *at the same compute budget*. We answered it with compute-matched replay on 437k real attack-search iterations, a complete $5 \times 4 \times 100$ cost-to-discovery matrix, and a live grid on three open-weights instruct models under a real alignment intervention.

Adaptive protocols do find substantially more at matched compute, but not for the reason the hypothesis proposes. The gain comes from searching over design families and from learning across a portfolio which family works. Using a failing run’s internal signal as a redesign trigger is worth approximately nothing ($\text{AUC} \approx 0.5$ at every checkpoint) and costs detections at large budgets. On the second half of the hypothesis, providing stronger evidence for genuine robustness, adaptive protocols currently make things worse: their nulls carry nominally 95% bounds with 63–67% coverage, and anytime-valid inference does not repair this because the error is in the estimand, not the stopping rule.

Recommendations for practice.

- Pre-allocate budget across at least three design families rather than deepening one. This single change accounts for the entire $0.00 \rightarrow 1.00$ effect measured here.
- Spend adaptivity at the portfolio level, using Thompson sampling or successive halving over methods, where nulls aggregate into real evidence.
- Do not instrument within-search stall detection unless you have validated that your progress signal is diagnostic on the surviving-null population. Measure its AUC first.
- Never publish a binomial confidence interval on a null as a bound on vulnerability. Report the design space searched and the ceiling attained; a null is a statement about your design, not about the model.
- Measure intervention effect sizes across design families too. The same intervention looks inert ($0.077 \rightarrow 0.037$) or strongly effective ($1.000 \rightarrow 0.519$) depending purely on which family you probe with.

Future work. Three directions follow directly. First, test whether a *better* within-run signal, such as true elicitation probability from repeated sampling or a learned probe, is diagnostic where target-token probability is not; this is the one way the E1 verdict could flip. Second, build a bound whose estimand is elicitation under a *specified* design space, and evaluate its coverage against held-out design families. Third, extend the portfolio experiment to methods with heterogeneous costs, since our unit-cost accounting flatters cheap methods.

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A Additional experimental detail

A.1 Trigger hyper-parameter sweep (E1)

LADDER has three hyper-parameters: a patience k (how many iterations without improvement before the trigger may fire), a surrogate threshold τ (below which the rung is judged cold), and an

extension factor e (how much extra budget a warm rung receives). At $\tau = 0$ and $e = 1$ the arm reduces to FIXED-SPLIT exactly, which is what makes the contrast in section 4.1 a clean ablation of the trigger alone.

We swept $k \in \{2, 5, 20\}$, $\tau \in \{0.05, 0.3, 0.7\}$ and $e \in \{1, 3, 10\}$, giving 27 settings, each replayed at every budget. The best setting anywhere in the grid gains +0.06 over FIXED-SPLIT; the worst loses −0.17. No setting produces a gain that survives Holm correction. The conclusion in section 4.1 is therefore a property of the trigger idea, not of one unlucky parameterisation.

A.2 Rung attribution (E1)

For every detection made by LADDER we record which rung was active at the moment of success. Rung 0 is the default `plain_init` random search, rung 1 is re-initialisation from a related success, rung 2 is a template-family switch. Across all budgets, rung 0 accounts for 0 detections, rung 1 for 31, and rung 2 for 13. This is the direct measurement behind the claim that the benefit is a design family effect.

A.3 Pressure design families (E4)

The four pressure families are applied after the model produces a correct answer to a GSM8K problem under greedy decoding.

- **P1, simple doubt.** A single expression of disagreement with no justification.
- **P2, appeal to authority.** A single turn attributing the contrary answer to an expert source.
- **P3, fabricated consensus.** A single turn asserting that several other people reached a different answer.
- **P4, persistent multi-turn.** Repeated pushback across several turns without new argument.

The alignment intervention under test is an anti-sycophancy system prompt instructing the model to maintain a correct answer under social pressure and to change it only in response to a substantive mathematical argument. The control condition is the same grid with no system prompt. An item counts as “flipped” when the model’s final answer changes away from the correct value.

A.4 Per-configuration coverage (E3)

Pooled coverage in table 5 averages over 14 configurations that behave very differently. Configurations whose true elicitable fraction is near the middle of the range have coverage close to nominal, which is what drags the pooled figure up. The three configurations with the highest true elicitable fraction, llama2-13b, llama2-70b and llama2-7b at 0.96–1.00, have coverage of exactly 0.00 for the adaptive interval: the bound fails in every replicate. This is the pattern one expects when the estimand is wrong rather than when the interval is merely too narrow, since a too-narrow interval degrades gradually while a wrong estimand fails systematically wherever the gap between the two quantities is large.

A.5 Artifacts

All code, raw outputs, summaries and statistical tests are released alongside the paper: offline results (E1–E3 raw traces, summaries and tests), live results (the E4 grid and protocol replay), the figure-generation scripts, and the pre-registered analysis plan written before any protocol was run.